

ACADEMIC DETAILS				
Degree	Specialization	Institute	Year	CPI/%
B.Tech.	Computer Science & Engineering	IIT Gandhinagar	2024–Present	8.80
Class XII	Physics, Chemistry, Maths	Modern Vidya Niketan, Sec-17, Faridabad	2022–2024	95.4
Class X		Modern Vidya Niketan, Sec-17, Faridabad	2012–2022	96.8

RELEVANT COURSEWORK

- Computer Science:** Data Structures and Algorithms, Theory of Computing, Discrete Mathematics
- Artificial Intelligence:** Probability, Statistics & Data Visualization, Machine Learning, Introduction to Data Science
- Mathematics:** Calculus of Single Variable & Linear Algebra, Calculus of Several Variables, Ordinary Differential Equations

PROJECTS

- Fairness in Federated Learning: Client-Side Interventions, Server Protocols & Adaptive Weighting**

Advisor: Prof. Manisha Padala | [PyTorch](#), [CNNs](#), [Federated Learning](#) | [GitHub](#) [Jan '26 – Apr '26]
 - Developed a federated learning framework to analyze and mitigate demographic bias in gender classification using UTKFace and FairFace datasets.
 - Implemented Dynamic Client-Adaptive Fairness Weighting (DCA-FW), reducing the race-wise fairness gap from 8.5% to 5.1% while maintaining model accuracy.
 - Evaluated adversarial fairness and label-flipping attack strategies to study robustness and bias mitigation in distributed learning systems.
- Under Vehicle Surveillance System, IIT Gandhinagar**

Advisor: Prof. Shanmuganathan Raman | [Live Demo](#) [Aug '25 – Nov '25]
 - Implemented omnidirectional camera calibration using the Scaramuzza OCamCalib model — building perspective and panoramic undistortion lookup tables consumed by OpenCV's `remap()` — to correct barrel distortion in fisheye undercarriage imagery.
 - Designed a three-stage pipeline (undistortion → config generation → stitching) integrating a C++ OpenPano executable with SIFT, RANSAC homography, bundle adjustment, and multi-band blending; deployed as a live Streamlit app.
- Semantic Search Engine: Dense Retrieval & Hybrid Ranking | Python, C++, BERT, FAISS**

Advisor: Prof. Anirban Das Gupta | [GitHub](#) | [Live Demo](#) [Jan '26 – Apr '26]
 - Improved Recall@10 from ~0.72 to ~0.95 on MS MARCO by designing a hybrid ranking pipeline fusing BM25 and BERT-encoded dense retrieval scores via a tunable α parameter.
 - Reduced query latency by 32% (8.4 ms → 5.7 ms) on a 100k-document corpus by replacing exhaustive L2 search with HNSW and FAISS ANN indexing via a custom C++ index with pybind11 bindings.
 - Boosted top-k precision on SciFact with a cross-encoder reranker using joint [CLS] representations; deployed as an interactive Streamlit interface for real-time model comparison.
- RAG AI Teaching Assistant | Python, Whisper, LLMs, RAG, Embeddings**

[GitHub](#) | [Live Demo](#) [Dec '25]
 - Built an end-to-end Retrieval-Augmented Generation (RAG) system to answer questions from lecture videos by converting video content into a searchable knowledge base.
 - Implemented transcript chunking, embedding generation, cosine-similarity retrieval, and context-aware response generation using Whisper and GPT/DeepSeek models.
 - Developed a retrieval pipeline over hours of lecture content, enabling semantic search and grounded question answering from custom datasets.

TECHNICAL SKILLS

- Languages:** Python, C/C++
- Tools & Technologies:** Git, GitHub, Canva, Docker
- Libraries:** SentenceTransformers, FAISS, Streamlit, NumPy, Pandas, Matplotlib, scikit-learn
- Frameworks (familiar):** PyTorch, TensorFlow
- Foundations:** Gradient descent, backpropagation, cross-entropy loss, regularization, Bayesian inference, probabilistic modelling, financial modelling

ACHIEVEMENTS

- AIR 1836** in JEE Advanced among 1,80,000 appeared candidates
- AIR 6362** (99.6 percentile) in JEE Mains among 14 lakh+ appeared candidates
- Codeforces** Max Rating **1231** (Pupil); solved **400+** DSA problems across LeetCode, Codeforces, and TUF

- Silver Medal in Basketball League, IIT Gandhinagar; represented school multiple times in district-level competitions
- Selected for the **IIT Gandhinagar Explorer Fellowship**, completing a fully sponsored 42-day exploration across multiple Indian states

POSITIONS OF RESPONSIBILITY

- **Transportation Coordinator, Welfare Council, IIT Gandhinagar** [May '26 – Present]
 - Managing all transportation-related operations on campus, including bus scheduling, cycle allocation, and student feedback collection in coordination with the Welfare Council.
- **Student Guide, IIT Gandhinagar** [Apr '26 – Present]
 - Mentoring 10 incoming students to help them acclimatize to the academic and social environment at IIT Gandhinagar.
- **Finance Member, Sports Council, IIT Gandhinagar** [Aug '25 – May '26]
 - Contributed to the organization of sports council events and managed finance-related tasks as required.
- **Marketing Senior Team Member, Blithchron (Cultural Fest), IIT Gandhinagar** [Apr '25 – Mar '26]
 - Conducted outreach visits to colleges to promote Blithchron events; additionally handled digital marketing campaigns for the fest.